

# FDE Xlerate

Week 1, Day 4

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# The problem

## ■ Task

- Build a real PoC - a customer service agent that answers account questions by answering a CRM
- In agent over system of record engagements, the client won't provide access to the real data. The FDE must build the agent against mock data while developing the correct infrastructure such that 'swapping in the real system is a configuration change and not a rewrite.'

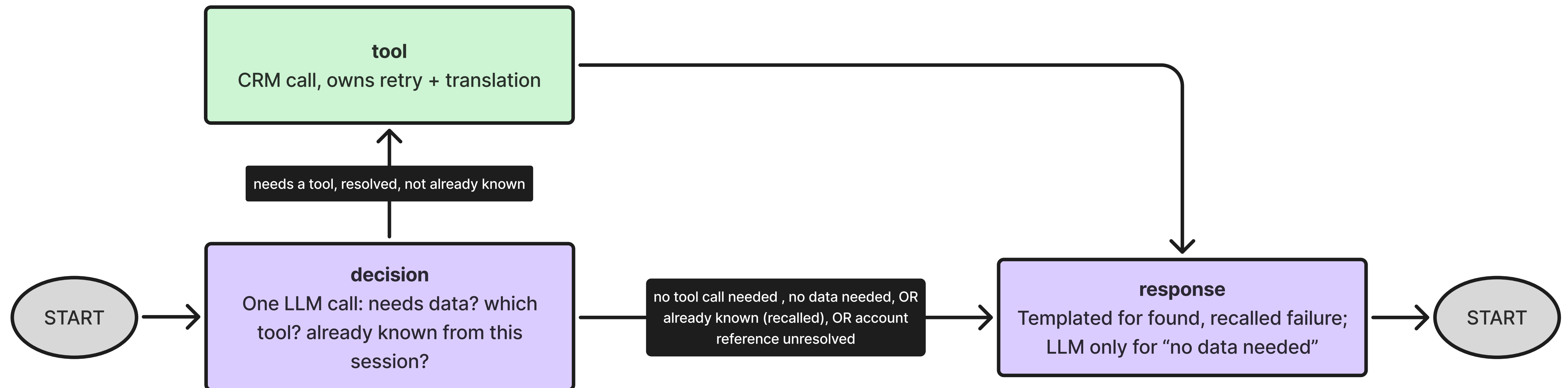
## ■ Design doc

- The design doc is as high priority as the code itself - this is what the client will read and decide on before continuing with the engagement

## ■ Business problem

- Industry: banking
- Case: a customer-service agent that answers bank-account questions by calling a CRM
- Priorities
  - Minimize calling the CRM, ensuring it's only queried when there is requisite information the agent doesn't already have
  - The agent must never invent data. If querying the CRM returns nothing, the agent must say so plainly and offer escalation

# High-level flow



# Architecture

## Decision

- **decision.py**
- One LLM call per turn, three-shape
  - contract
    - answer\_directly
    - call\_tool
    - recall\_from\_history
- Classifies how an account was referenced rather than naming an ID itself. Classifications include:
  - explicit
  - last\_mentioned
  - the\_other\_one
  - none
- recall\_from\_history names the same tool and account\_reference a fresh call\_tool would
  - A miss falls back to a real call rather than guessing.

## Conversation-state

- **conversation\_state.py**
- Pure, LLM-free turn history plus two resolvers:
  - resolve\_other\_account()
  - resolve\_recall()
  - both return None on ambiguity rather than guess.

## Tool layer

- **tool\_client.py**
- The hard boundary.
- Owns HTTP, timeouts, and retry
- Collapses every failure into found, not\_found or unavailable
- Swapping in a real CRM later is a config change, not a rewrite.

# Architecture

## Response

- **response.py**
- Templated for found/recalled/failure, LLM only for "no data needed"
- Zero room to improvise on the graded path

## Graph

- **graph.py**
- Decision → tool → response
- No loop-back; the only retry lives inside the tool client

## Mock CRM

- Standalone FastAPI service, two independent fault-injection paths:
  - A header override (tests)
  - Fixture-flagged accounts (the live demo).

# Tech Stack

## ■ Mock CRM

FastAPI + uvicorn, fixture data, header-driven fault override plus fixture-flagged demo accounts. Contract in mock\_crm/contract.md

## ■ Orchestration

LangGraph StateGraph (langgraph==1.2.11, same pin as day 3), with InMemorySaver checkpointing

## ■ Tests

pytest — 43 tests, LLM mocked throughout, real HTTP against a live uvicorn mock for the tool/graph layers, under 2s

## ■ Tool layer

httpx, explicit timeouts, hand-rolled retry (3x, exponential backoff + jitter)

## ■ Model/ SDK

claude-sonnet-5 via anthropic==0.104.1, same pin carried from days 2–3